# 20 Lecture 20

## 20.1 Central Limit Theorem

The strong law of large numbers states that the average of $n$ independent and identically distributed random variables, $S_n/n$, for large $n$ is very close to their common mean value $\mu := \mathbb{E}[X]$. Now, with the additional assumption $\sigma := \mathrm{Var}(X) \in (0, \infty)$, the *Central Limit Theorem* (CLT) specifies how close these two (random) quantities are. It states that we can write

$$\frac{S_n}{n} = \mu + \frac{\sigma}{\sqrt{n}} Z,$$

with $Z$ approximately following the standard normal distribution $N(0, 1)$. Equivalently,

$$S_n = n\mu + \sigma\sqrt{n}\, Z,$$

that is, typically $S_n$ lies at a distance of order $\sqrt{n}$ around its mean value $n\mu$. This is the phenomenon, initially observed by De Moivre and Laplace, we described in Lecture 7 (Example 7).

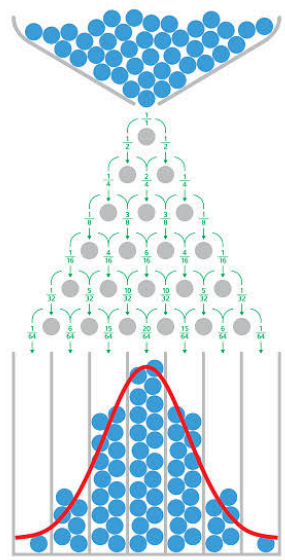

Figure 1: Galton board illustrating the emergence of the normal distribution: the horizontal position of each ball follows a binomial distribution, being the sum of independent Bernoulli trials.

Only For 525 Winter 2026
No Permission For Uploading
Has Been Given

**Theorem 20.1** (Lindeberg–Lévy CLT)**.** *Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of i.i.d random variables with $\mathbb{E}[X_1] = \mu$ and $Var(X_1) = \sigma^2 \in (0, +\infty)$. For $n \in \mathbb{N}$, we set $S_n = X_1 + \ldots + X_n$. Then*

$$\frac{S_n - n\mu}{\sqrt{n\sigma^2}} \xrightarrow{d} Z, \text{ where } Z \sim N(0, 1).$$

**Remark 20.2.** The central limit theorem provides approximations for probabilities involving sums of independent and identically distributed random variables. It has applications in Statistics, for example in the construction of approximate confidence intervals and in hypothesis testing. We use it by considering a probability of the form $\mathbb{P}\left(\frac{S_n - n\mu}{\sqrt{n\sigma^2}} \in A\right)$.

**Example 1**
We have a coin and want to test whether it is fair. We toss it 100 times, and "Heads" comes up 38 times. Should we start to consider that it might not be fair?

We consider the random variables $(X_n)_{n\in\mathbb{N}}$ with $X_n = 1$ if the the $n$-th toss is Heads and $X_n = 0$ if the $n$-th toss is tails. The total number of Heads after $n$ tosses is $S_n := X_1 + \ldots + X_n$. Let's assume that the coin is fair and lets compute $\mathbb{P}(S_{100} \leq 38)$. We have $\mu = \mathbb{E}[X_1] = 1/2$ and $\sigma^2 = \mathrm{Var}(X_1) = p(1-p) = 1/4$, therefore by CLT

$$\mathbb{P}(S_{100} \leq 38) = \mathbb{P}\left(\frac{S_{100} - 100 \cdot 0.5}{\sqrt{100/4}} \leq \frac{38 - 100 \cdot 0.5}{\sqrt{100/4}}\right) \approx \mathbb{P}(Z \leq -12/5) = \Phi(-2.4) \approx 0.0082.$$

The probability of $S_{100} \leq 38$ is smaller that 1%, therefore we should start considering the possibility that the coin is not fair.

If we had Heads 43 times, the same calculation yields the estimate $\mathbb{P}(S_{100} \leq 43) \approx 0.0887$. This is almost 9% and in this case we would not be as suspicious.

**Example 2**
We consider the breaking strength of a specific wire, which is a random variable with unknown mean $\mu$ and variance $\sigma^2 = 1/10$. We want to determine approximately the mean $\bar{\mu}$ from a very large quantity of wire just produced by a factory. We perform $n$ independent measurements $X_1, X_2, \ldots, X_n$ and will use the average

$$\bar{X}_n := \frac{X_1 + \cdots + X_n}{n}$$

as an approximation of $\mu$. How large should $n$ be so that $|\bar{X}_n - \mu| \leq 1/100$ with probability at least 0.95?

For the probability of the event of interest, we use the following approximation:

$$\mathbb{P}\big(|S_n/n - \mu| \leq 1/100\big) = \mathbb{P}\big(|S_n - n\mu|/\sqrt{n\sigma^2} \leq \sqrt{n}/100\sigma\big) \approx \mathbb{P}\big(|Z| \leq \tfrac{\sqrt{n}}{100\sigma}\big),$$

where $Z$ is a $N(0,1)$ random variable. Hence,

$$\mathbb{P}\big(|Z| \leq \tfrac{\sqrt{n}}{100\sigma}\big) = \Phi\Big(\tfrac{\sqrt{n}}{100\sigma}\Big) - \Phi\Big(-\tfrac{\sqrt{n}}{100\sigma}\Big) = 2\Phi\Big(\tfrac{\sqrt{n}}{100\sigma}\Big) - 1.$$

The last quantity is at least 0.95 if

$$\Phi\Big(\tfrac{\sqrt{n}}{100\sigma}\Big) \geq 1 - 0.025.$$

We can find the unique number $z_{0.025}$ such that $\Phi(z_{0.025}) = 1 - 0.025$. So

$$\sqrt{n}\frac{1}{100\sigma} \geq z_{0.025} \approx 1.96 \implies n \geq (100\sigma z_{0.025})^2 \approx 384.16.$$

**Remark 20.3.** We note that in order to approximate $\mathbb{P}\big(|S_n - n\mu|/\sqrt{n\sigma^2} \leq \sqrt{n}/100\sigma\big)$ in the above calculation can be justified as follows. By CLT we know that

$$\lim_{k\to\infty} \mathbb{P}\big(|S_k - k\mu|/\sqrt{k\sigma^2} \leq \sqrt{n}/100\sigma\big) = \mathbb{P}(|Z| \leq \sqrt{n}/100\sigma),$$

therefore for $k = n$ (assuming that $n$ is large enough), the numbers

$$\mathbb{P}\big(|S_n - n\mu|/\sqrt{n\sigma^2} \leq \sqrt{n}/100\sigma\big),\ \mathbb{P}(|Z| \leq \sqrt{n}/100\sigma)$$

will be close.

**Example 3**
The number of typographical errors on a page of a certain newspaper follows a Poisson distribution with mean $\lambda = 0.7$. If the newspaper has 64 pages, approximate the probability that at most 36 pages contain no errors at all. It is given that $e^{-0.7} \approx 1/2$, $\Phi(1) = 0.8413$, $\Phi(2) = 0.9773$, and $\Phi(3) = 0.9987$.

Only For Math 525 Winter 2026
No Permission For Uploading
Has Been Given

Let $Y_i$ be the number of typographical errors on page $i$. Define

$$X_i := \begin{cases} 1, & \text{if page } i \text{ has no typographical errors,} \\ 0, & \text{otherwise.} \end{cases}$$

We know that $Y_i \sim \text{Poisson}(0.7)$. Hence, $X_i$ follows a Bernoulli distribution with parameter

$$p = \mathbb{P}(Y_i = 0) = e^{-0.7}\frac{0.7^0}{0!} = e^{-0.7} \approx \tfrac{1}{2}.$$

Therefore,

$$\mathbb{E}[X_1] = p \approx \tfrac{1}{2}, \qquad \text{Var}(X_1) = p(1-p) \approx \tfrac{1}{4}.$$

Let $S_{64} = \sum_{i=1}^{64} X_i$ be the number of pages with no errors. Then

$$\mathbb{P}(S_{64} \le 36) \approx \mathbb{P}(S_{64} - 32 \le 4) = \mathbb{P}\left(\frac{S_{64} - 32}{\sqrt{64}/4} \le 2\right) \approx \Phi(2).$$

The first equality is an approximation because we used approximate values for $\mathbb{E}[X_1]$ and $\text{Var}(X_1)$. The final approximation follows from the Central Limit Theorem.

We finally present (without proof) a result that quantifies how good an approximation is $\Phi(x) = \mathbb{P}(Z \le x)$ for $\mathbb{P}(\frac{S_n - n\mu}{\sqrt{n\sigma^2}} \le x)$.

**Theorem 20.4** (Berry-Esseen theorem)**.** *Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of i.i.d random variables such that $\mu = \mathbb{E}[X_1]$, $\mathbb{E}[|X_1 - \mu|^2] = \sigma^2$ and $\mathbb{E}[|X_1 - \mu|^3] = \rho < \infty$. Then.*

$$\left|\mathbb{P}\left(\frac{S_n - n\mu}{\sqrt{n\sigma^2}} \le x\right) - \Phi(x)\right| \le \frac{3\rho}{\sigma^2\sqrt{n}}, \quad \textit{for any } x \in \mathbb{R}.$$

**Example 4**
A factory produces electronic components. Each component is either *defective* or *non-defective*. Let

$$X_i = \begin{cases} 1, & \text{if the } i\text{th tested component is defective,} \\ 0, & \text{otherwise.} \end{cases}$$

Assume that $(X_i)_{i\ge1}$ are independent and identically distributed with $\mathbb{P}(X_i = 1) = p$, where the defect probability $p$ is unknown. The manufacturer claims that the production process is under control, meaning that $H_0 : p \le 0.02$. To test this claim, an inspector tests $n$ randomly selected components and computes the sample proportion

$$\hat{p}_n := \frac{1}{n}\sum_{i=1}^{n} X_i.$$

Question: The inspector wants to estimate $p$ with an absolute error at most 0.005 and confidence at least 99%, that is, $\mathbb{P}\big(|\hat{p}_n - p| > 0.005\big) \le 0.01$. How big should $n$ be?

We note that since $\mathbb{E}[X_i] = p$ and $\text{Var}(X_i) = p(1-p)$, we have

$$\mathbb{E}[\hat{p}_n] = p, \qquad \text{Var}(\hat{p}_n) = \frac{1}{n^2}\sum_{i=1}^{n}\text{Var}(X_i) = \frac{p(1-p)}{n}.$$

**1st way: Chebyshev's inequality.** By Chebyshev's inequality,

$$\mathbb{P}\big(|\hat{p}_n - p| \ge \varepsilon\big) \le \frac{\text{Var}(\hat{p}_n)}{\varepsilon^2} = \frac{p(1-p)}{n\varepsilon^2}.$$

Since $p(1-p) \le 1/4$ for all $p \in [0,1]$, we obtain $\mathbb{P}\big(|\hat{p}_n - p| \ge \varepsilon\big) \le \frac{1}{4n\varepsilon^2}$. We want this probability to be at most 0.01 with $\varepsilon = 0.005$, hence $\frac{1}{4n(0.005)^2} \le 0.01$. Solving for $n$ gives

$$n \ge \frac{1}{4 \cdot 0.01 \cdot (0.005)^2} = 1{,}000{,}000.$$

Only for Math 525 Winter 2026
No Permission for Uploading
Has Been Given

**2nd way: Hoeffding's inequality.** Since $0 \le X_i \le 1$, Hoeffding's inequality yields

$$\mathbb{P}\big(|\hat{p}_n - p| \ge \varepsilon\big) \le 2e^{-2n\varepsilon^2}.$$

We require

$$2e^{-2n(0.005)^2} \le 0.01,$$

which is equivalent to

$$e^{-2n(0.005)^2} \le 0.005.$$

Taking logarithms,

$$-2n(0.005)^2 \le \ln(0.005),$$

so

$$n \ge \frac{|\ln(0.005)|}{2(0.005)^2} \approx 106{,}000.$$

**3rd way: CLT with Berry–Esseen theorem.** Assume now that $p = 0.02$. Then

$$\mathbb{E}[X_1] = p, \qquad \sigma^2 := \mathrm{Var}(X_1) = p(1-p) = 0.0196, \qquad \sigma = 0.14.$$

Moreover, since $|X_1 - p| \le 1$, we have

$$\rho := \mathbb{E}\big[|X_1 - p|^3\big] < \infty,$$

so the Berry–Esseen theorem applies. We want

$$\mathbb{P}\big(|\hat{p}_n - p| \le \varepsilon\big) = \mathbb{P}\big(|S_n - np| \le n\varepsilon\big) \ge 0.99, \qquad \varepsilon = 0.005.$$

Equivalently,

$$\mathbb{P}\left(\left|\frac{S_n - np}{\sigma\sqrt{n}}\right| \le \frac{\varepsilon\sqrt{n}}{\sigma}\right) \ge 0.99.$$

By the Berry–Esseen theorem, for all $x > 0$,

$$\mathbb{P}\left(\left|\frac{S_n - np}{\sigma\sqrt{n}}\right| \le x\right) \ge 2\Phi(x) - 1 - \frac{6\rho}{\sigma^2\sqrt{n}}.$$

Hence it suffices to choose $n$ such that

$$2\Phi\left(\frac{\varepsilon\sqrt{n}}{\sigma}\right) - 1 - \frac{6\rho}{\sigma^2\sqrt{n}} \ge 0.99.$$

Ignoring the error term for a first approximation, we require

$$\frac{\varepsilon\sqrt{n}}{\sigma} \ge z_{0.995} \approx 2.576,$$

which yields

$$\sqrt{n} \ge \frac{2.576\,\sigma}{\varepsilon} = \frac{2.576 \cdot 0.14}{0.005} \approx 72.1, \qquad n \ge 5200.$$

Thus, a sample size of approximately $n \approx 5{,}200$ trials is sufficient to guarantee

$$\mathbb{P}\big(|\hat{p}_n - p| \le 0.005\big) \ge 0.99.$$

**Comparison.**

- Chebyshev's inequality gives an extremely conservative bound.
- Hoeffding's inequality is non-asymptotic and significantly sharper.
- The CLT with Berry–Esseen yields an even smaller required sample size.

Only For Math 535 Winter 2026
No Permission For Uploading
Has Been Given

## 20.2 Proof of CLT

### 20.2.1 Lévy continuity theorem

Notice that the statement of the central limit theorem involves convergence in distribution. We have seen two ways to prove convergence in distribution; namely Definition 18.1 and Proposition 18.3. Our proof will use a third one according to which

$$\text{The convergence } X_n \xrightarrow{d} X \Leftrightarrow \phi_{X_n}(t) \xrightarrow{n\to\infty} \phi_X(t), \text{ for any } t \in \mathbb{R}.$$

Each of these three methods works well in different cases. The first is useful in situations where the distribution function of $X_n$ is easy to compute (e.g., if $X_n$ concerns the maximum or minimum of a sequence of random variables or has a known distribution). The third is useful when $X_n$ involves a sum of independent random variables (such is the case of the central limit theorem).

**Theorem 20.5** (Lévy continuity theorem)**.** *Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of random variables and $(\phi_{X_n})_{n\in\mathbb{N}}$ be the sequence of their characteristic functions.*

1. *If $X$ is a random variable such that $X_n \xrightarrow{d} X$, then $\lim_{n\to\infty} \phi_{X_n}(t) = \phi_X(t)$.*
2. *If the limit $\lim_{n\to\infty} \phi_{X_n}(t) =: f(t)$ exists for all $t \in \mathbb{R}$ and the function $f$ is continuous at $0$, then there exists a random variable $X$ such that $\phi_X(t) = f(t)$ and $X_n \xrightarrow{d} X$.*

**Corollary 20.6.** *Let $(X_n)_{n\in\mathbb{N}}, X$ be random variables with values in $\mathbb{R}$. The, $X_n \xrightarrow{d} X$ if and only if $\phi_{X_n}(t) \xrightarrow{n\to\infty} \phi_X(t)$, for any $t \in \mathbb{R}$.*

The following Lemma is useful, when we try to compute limits of the form $\lim_{n\to\infty} \phi_{X_n}(t)$.

**Lemma 20.7.** *Let $(c_n)_{n\in\mathbb{N}}$ be a sequence in $\mathbb{C}$ such that $\lim_{n\to\infty} nc_n = a \in \mathbb{C}$. Then,*

$$\lim_{n\to\infty} (1 + c_n)^n = e^a.$$

*Proof.* For $n \in \mathbb{N}$, we observe that $\sum_{j=1}^n |c_n|^2 = n|c_n|^2 = |nc_n|^2/n \xrightarrow{n\to\infty} 0$. This implies $|c_n| \le 1/2$ for any $n$ large enough and hence $1 + c_n \notin (-\infty, 0]$. We have

$$\frac{\prod_{j=1}^n (1 + c_n)}{e^{\sum_{j=1}^n c_n}} = e^{\sum_{j=1}^n (\log(1+c_n) - c_n)}.$$

The exponent of the right hand side can be bounded by using the Taylor expansion of $\log(1 + z) = \sum_{k=1}^{+\infty} \frac{(-1)^{k-1}}{k} z^k$:

$$\left|\sum_{j=1}^n (\log(1 + c_n) - c_n)\right| \le \sum_{j=1}^n |\log(1+c_n) - c_n| \le n \left|\sum_{k=2}^{+\infty} \frac{(-1)^{k-1}}{k} c_n^k\right| \le \frac{n}{2} \sum_{k=2}^{+\infty} |c_n|^k = \frac{n|c_n|^2}{2} \frac{1}{1 - |c_n|} \le n|c_n|^2,$$

which, as we saw, converges to 0 as $n \to \infty$. Thus,

$$\frac{(1 + c_n)^n}{e^{nc_n}} = \frac{\prod_{j=1}^n (1 + c_n)}{e^{\sum_{j=1}^n c_n}} \xrightarrow{n\to\infty} 1 \implies (1 + c_n)^n \xrightarrow{n\to\infty} e^a.$$

□

**Example 5**
Suppose that $X_n \sim \text{Bin}(n, \lambda/n)$ for all $n \in \mathbb{N}$ and $X \sim \text{Pois}(\lambda)$, for some $\lambda > 0$.

We have for any $t \in \mathbb{R}$

$$\phi_{X_n}(t) = \mathbb{E}[e^{itX_n}] = \sum_{k=0}^n e^{itk} \binom{n}{k} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} = \sum_{k=0}^n \binom{n}{k} \left(\frac{e^{it}\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k}$$

Only For Math 525 Winter 2026
Not Permission For Uploading
Has Been Given

$$= \left(1 - \frac{\lambda}{n} + \frac{\lambda e^{it}}{n}\right)^n \xrightarrow{n\to\infty} e^{\lambda(e^{it}-1)}.$$

We notice that $e^{\lambda(e^{it}-1)} = \phi_X(t)$, therefore $X_n \xrightarrow{d} X$.

### 20.2.2 Proof

**Lemma 20.8.** *Let $x \in \mathbb{R}$. The following inequality holds*

$$\left|e^{ix} - 1 - ix - \frac{1}{2}(ix)^2\right| \leq x^2,$$

*where $|\cdot|$ is the distance of a complex number from the origin.*

*Proof.* We observe

$$\left|e^{ix} - 1 - ix - \frac{1}{2}(ix)^2\right| = \left|i\int_0^x (e^{it} - 1 - it)dt\right| = \left|i^2\int_0^x\int_0^t (e^{is} - 1)dsdt\right|$$
$$\leq \int_0^x\int_0^t |e^{is} - 1|dsdt \leq 2\int_0^x\int_0^t dsdt = x^2.$$

□

**Lemma 20.9.** *Let $X$ be a random variable with values in $\mathbb{R}$ such that $\mathbb{E}[X^2] < \infty$. We set $\mu = \mathbb{E}[X]$ and $\mathbb{E}[X^2] = \beta$. Then*

$$\phi_X(t) = 1 + it\mu - \frac{t^2\beta}{2} + \nu(t),$$

*where $\nu$ satisfies $\lim_{t\to 0} \frac{\nu(t)}{t^2} = 0$.*

*Proof.* We will show that $\nu(t) = \phi_X(t) - (1 + it\mu - \frac{t^2\beta}{2})$ satisfies $\lim_{t\to 0} \frac{\nu(t)}{t^2} = 0$. Set $A(x) := e^{ix} - 1 - ix - (ix)^2/2$, for $x \in \mathbb{R}$. By the definition of $\mu, \beta$ we have

$$|\nu(t)| = \left|\phi_X(t) - (1 + it\mu - \frac{t^2\beta}{2})\right| = \left|\mathbb{E}\left[e^{itX} - 1 - itX - \frac{(itX)^2}{2}\right]\right| \leq \mathbb{E}[|A(tX)|],$$

hence $\frac{|\nu(t)|}{t^2} \leq \frac{\mathbb{E}[|A(tX)|]}{t^2}$. We pick any sequence $(t_n)_{n\in\mathbb{N}} \in \mathbb{R}\setminus\{0\}$ with $t_n \xrightarrow{n\to\infty} 0$ and we set $Y_n := |A(t_nX)|/t_n^2$. Then,

- $|Y_n| \leq (t_nX)^2/t_n^2 = X^2$ by Lemma 20.6 and $\mathbb{E}[X^2] < \infty$.
- By Taylor's theorem
$$Y_n = \left|\sum_{k=3}^{+\infty} \frac{(it_nX)^k}{k!}\frac{1}{t_n^2}\right| = \left|\sum_{k=3}^{+\infty} t_n^{k-2}\frac{(iX)^k}{k!}\right| \xrightarrow{n\to\infty} 0.$$

By the Dominated Convergence Theorem we get $\lim_{n\to\infty}\mathbb{E}[Y_n] = \mathbb{E}[\lim_{n\to\infty} Y_n] = 0$. The result follows. □

*Proof of CLT.* We compute the characteristic function of $Z_n = \frac{S_n - n\mu}{\sqrt{n\sigma^2}}$. For $t \in \mathbb{R}\setminus\{0\}$ we have

$$\phi_{Z_n}(t) = \mathbb{E}\left[e^{it\frac{X_1-\mu}{\sqrt{n\sigma^2}}}\dots e^{it\frac{X_n-\mu}{\sqrt{n\sigma^2}}}\right] = \mathbb{E}\left[e^{it\frac{X_1-\mu}{\sqrt{n\sigma^2}}}\right]\dots\mathbb{E}\left[e^{it\frac{X_n-\mu}{\sqrt{n\sigma^2}}}\right] = \phi_{X_1-\mu}(t/\sqrt{n\sigma^2})\dots\phi_{X_n-\mu}(t/\sqrt{n\sigma^n})$$
$$= \left(\phi_{X_1-\mu}\left(t/\sqrt{n\sigma^2}\right)\right)^n.$$

Only For Math 525 Winter 2026
No Permission For Uploading
Has Been Given